

RARITONE DATA ANALYST INTERNSHIP

FINAL PROJECT REPORT

Submitted By

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Internship Role

Data Analyst Intern

Internship Organization

Raritone Private Limited

Duration

1 Month

ABSTRACT

The fashion e-commerce industry is rapidly evolving with increasing reliance on data-driven decision-making and personalized user experiences. Customers now expect intelligent recommendations, seamless navigation, and virtual product visualization before making purchase decisions.

This internship project focuses on analyzing customer behavior, engagement patterns, retention metrics, conversion funnels, and virtual try-on interactions within a fashion e-commerce ecosystem.

The objective of the study was to understand how data analytics can improve user engagement, increase customer retention, reduce cart abandonment, and enhance overall business performance.

The project includes:

- KPI framework development
- Customer segmentation analysis
- Engagement and retention analysis
- Conversion funnel analysis
- Virtual try-on behavioral insights
- Competitor benchmarking
- Dashboard creation
- Business intelligence reporting
- Recommendation system research

Key findings indicate that personalization, AI-based recommendations, virtual try-on experiences, and optimized user journeys significantly enhance customer engagement, conversion rates, and retention performance.

CHAPTER 1: INTRODUCTION

1.1 Background

The fashion e-commerce industry has transformed significantly with digital adoption. Online platforms are no longer limited to product listings but now focus on immersive shopping experiences powered by data analytics and artificial intelligence.

Modern users expect:

- Personalized recommendations
- High-quality product visualization
- Fast and seamless shopping experience
- Accurate size and fit suggestions
- Interactive features like virtual try-on

Raritone aims to solve these challenges through a next-generation fashion technology platform.

1.2 About Raritone

Raritone is a virtual fashion platform designed to improve online shopping experiences using:

- Virtual Try-On Technology
- AI-powered personalization systems
- Customer behavior analytics
- Smart recommendation engines
- Digital wardrobe management

The platform focuses on reducing uncertainty in online shopping and improving user confidence before purchase.

1.3 Problem Statement

Traditional fashion e-commerce platforms face several limitations:

- Lack of physical product experience
- High return rates due to size mismatch
- Low personalization in recommendations
- Poor engagement retention
- Inefficient product discovery

These issues directly impact customer satisfaction and business revenue.

1.4 Need for Data Analytics

Data analytics plays a crucial role in solving these challenges by:

- Understanding customer behavior patterns
- Tracking user engagement

- Improving product recommendations
- Reducing churn rate
- Increasing conversion efficiency
- Supporting strategic decision-making

CHAPTER 2: PROJECT OBJECTIVES

The primary objectives of this internship were:

2.1 User Behavior Analysis

To understand how users interact with fashion e-commerce platforms.

2.2 KPI Identification

To define measurable metrics for performance tracking.

2.3 Engagement Optimization

To analyze factors affecting user engagement and retention.

2.4 Conversion Funnel Study

To identify drop-off points in the customer journey.

2.5 Virtual Try-On Impact Analysis

To evaluate how try-on features influence user decisions.

2.6 Competitor Benchmarking

To compare Raritone with leading fashion platforms.

2.7 Dashboard Development

To visualize key metrics for business decision-making.

2.8 Recommendation System Insights

To study how AI can improve personalization.

CHAPTER 3: TOOLS AND TECHNOLOGIES USED

3.1 Microsoft Excel

Used for:

- KPI calculations
- Data cleaning
- Pivot tables

- Graphical analysis
- Dashboard creation

3.2 Google Sheets

Used for:

- Collaboration
- Data sharing
- Real-time reporting

3.3 Power BI

Used for:

- Advanced dashboards
- Interactive visualizations
- Trend analysis

3.4 Google Analytics (Theoretical Study)

Used to understand:

- User behavior tracking
- Traffic sources
- Conversion analysis
- Retention monitoring

CHAPTER 4: KPI FRAMEWORK DESIGN

KPIs were categorized into four major groups:

4.1 User Growth KPIs

- Total Users
- New Users
- App Installs
- User Acquisition Rate

4.2 Engagement KPIs

- Daily Active Users (DAU)
- Monthly Active Users (MAU)

- Session Duration
- Product Views per Session
- Virtual Try-On Engagement Rate

4.3 Retention KPIs

- Day 1 Retention
- Day 7 Retention
- Monthly Retention
- Churn Rate

4.4 Conversion KPIs

- Add to Cart Rate
- Checkout Completion Rate
- Purchase Conversion Rate
- Funnel Drop-off Rate

CHAPTER 5: USER ENGAGEMENT ANALYSIS

5.1 Overview

User engagement analysis focuses on how frequently and effectively users interact with the platform.

5.2 Key Observations

- Personalized recommendations significantly improve engagement
- Visual content increases product interaction rates
- Virtual try-on users spend more time on platform
- Mobile devices dominate user traffic
- Peak usage occurs during evening hours

5.3 Insights

Engagement is strongly influenced by personalization, visual experience, and ease of product discovery.

CHAPTER 6: CUSTOMER BEHAVIOR ANALYSIS

6.1 Customer Journey

Awareness → Discovery → Product View → Try-On → Cart → Checkout → Purchase

6.2 Behavioral Factors

- Product aesthetics influence decision-making
- Pricing plays a key role in conversion
- Reviews increase trust
- Personalization improves engagement
- Ease of navigation improves retention

6.3 Key Insight

Customers are more likely to purchase when they experience personalization and visualization tools.

CHAPTER 7: RETENTION ANALYSIS

7.1 Importance

Retention directly impacts revenue, lifetime value, and business sustainability.

7.2 Retention Drivers

- Personalized recommendations
- Push notifications
- Loyalty programs
- Virtual try-on experience
- Re-engagement campaigns

7.3 Insight

Users engaged with personalization features show higher retention rates.

CHAPTER 8: CONVERSION FUNNEL ANALYSIS

8.1 Funnel Stages

App Open → Product View → Try-On → Add to Cart → Checkout → Purchase

8.2 Drop-off Points

- Product to Try-On skip
- Cart abandonment
- Checkout friction

8.3 Improvement Strategies

- Simplify checkout process
- Introduce one-click payments
- Offer cart reminders
- Provide dynamic discounts

CHAPTER 9: VIRTUAL TRY-ON ANALYTICS

9.1 Overview

Virtual Try-On allows users to visualize clothing before purchase using digital simulation.

9.2 Benefits

- Increased customer confidence
- Higher engagement time
- Improved conversion rate
- Reduced product returns

9.3 Future Enhancements

- AI-based body mapping
- Outfit recommendation system
- Size prediction engine
- Social sharing features

CHAPTER 10: COMPETITOR ANALYSIS

Myntra

Strong personalization and loyalty systems

AJIO

Competitive pricing and catalog variety

Nykaa Fashion

Premium branding and lifestyle focus

Amazon Fashion

Large inventory with lower personalization

CHAPTER 11: CUSTOMER SEGMENTATION

- New Users

- Active Users
- Frequent Buyers
- Inactive Users
- Try-On Enthusiasts
- High-Value Customers

Each segment requires different engagement strategies.

CHAPTER 12: DASHBOARD DEVELOPMENT

Dashboards were developed to track:

- User growth trends
- Engagement metrics
- Retention performance
- Conversion funnel
- Product performance
- Try-on analytics

Dashboards improve decision-making efficiency and visibility.

CHAPTER 13: BUSINESS RECOMMENDATIONS

- Improve AI-based recommendation engine
- Enhance virtual try-on accuracy
- Introduce loyalty programs
- Optimize checkout flow
- Improve personalization systems
- Strengthen product discovery

CHAPTER 14: CHALLENGES FACED

- Understanding analytics terminology
- Building KPI frameworks
- Designing dashboards
- Interpreting user behavior
- Working with simulated datasets

- Funnel analysis complexity
- Virtual try-on concept understanding

CHAPTER 15: KEY LEARNINGS

- Data Analytics fundamentals
- KPI tracking systems
- Customer behavior analysis
- Conversion optimization
- Retention strategies
- Dashboard development
- Business intelligence
- Fashion e-commerce analytics

CHAPTER 16: PROJECT OUTCOMES

- Defined KPIs for Raritone
- Built dashboards for analysis
- Studied customer behavior
- Identified funnel drop-offs
- Conducted competitor benchmarking
- Generated business insights
- Suggested improvements for growth

CONCLUSION

The Raritone Data Analyst Internship provided deep exposure to real-world data analytics in the fashion e-commerce domain.

The project improved understanding of customer intelligence, engagement analysis, retention strategies, and conversion optimization.

Overall, this internship strengthened analytical thinking, visualization skills, and business decision-making abilities, making it a highly valuable professional experience.